

A Philosophical-Mathematical Representation of Deductive and Inductive Logic: A Diagrammatic and Formal Analysis

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Preprint draft, October 7, 2026
Revised from the February 2025 version

Abstract

We give a diagrammatic representation of deductive and inductive reasoning in which deduction is a certain passage from thought to existence, $\varphi \rightarrow \exists$, and induction is a tentative, iterated passage from the study of existence to the theory of thought, $s\exists \circ t\varphi$. We make the diagram precise in a possible-worlds semantics: thoughts are propositions about worlds, deduction is semantic entailment, polarity $\uparrow, \downarrow$ distinguishes affirmation from modus tollens, and an inductive theory is supported by a study when some world is consistent with both. Within this setting we prove that deduction is sound and transitive, that iterated observation can only narrow the supported theories, that a single counterexample falsifies a universal theory, and that no finite study entails a universal theory over an unobserved individual. The last statement is a formal version of Hume’s problem of induction. Every theorem in Sections 4–6 is verified in the Lean 4 proof assistant; Appendix A lists the corresponding declarations.

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1 Introduction

Deductive logic derives particular conclusions from general premises with certainty: all men are mortal; Socrates is a man; therefore Socrates is mortal [1]. Inductive logic builds general theories from particular observations with only tentative support: many swans have been seen to be white, so perhaps all swans are white. Textual arguments and static diagrams, such as Venn diagrams and Peirce’s existential graphs [8], show the structure of a single inference well but do not show reasoning as a process in which deduction and induction alternate.

This paper proposes a diagram (Figure 1) that places both modes of reasoning in one picture, with a straight arrow for deduction, a circular arrow for induction, and a foundation of iterations, hypotheses, levels, binary states and polarities. We then give the diagram a precise semantics and prove what it asserts.

Contributions.

1. A diagrammatic framework for deduction $\varphi \rightarrow \exists$ and induction $s\exists \circ t\varphi$ (Section 2).
2. A possible-worlds semantics for every symbol of the diagram (Section 3).
3. Theorems for each mode: soundness, transitivity and modus tollens for deduction (Section 4); monotone refinement, falsification and the failure of induction as deduction (Theorem 5.4) for induction (Section 5); and a truth-value characterization of the foundation (Section 6).
4. A machine-checked formalization of all of these results in Lean 4 [2] (Appendix A).

2 The diagrammatic framework

The diagram has two primary rows and a foundation (Figure 1).

- **Deduction**, $\varphi \rightarrow \exists$. A thought φ (premises) leads to an existence $\exists$ (a conclusion) along a straight arrow, read *hence*. The passage is certain. Polarity $\uparrow$ marks an affirmed conclusion and $\downarrow$ a negated one.
- **Induction**, $s\exists \circ t\varphi$. The study of existence $s\exists$ (observations) leads to a theory of thought $t\varphi$ (a hypothesis) along a circular arrow, read *conclude*. The passage is tentative and is repeated as new observations arrive.
- **Foundation**. Iterations i , hypotheses h and levels l , together with binary states 0, 1 and polarities $\uparrow, \downarrow$, generate the thoughts used above.

Table 1 fixes the reading of each symbol. One change from the 2025 version: the symbol $\exists$ is kept for the diagram only, and the set of possible existences is written $\mathcal{W}$, so that it is not confused with the existential quantifier.

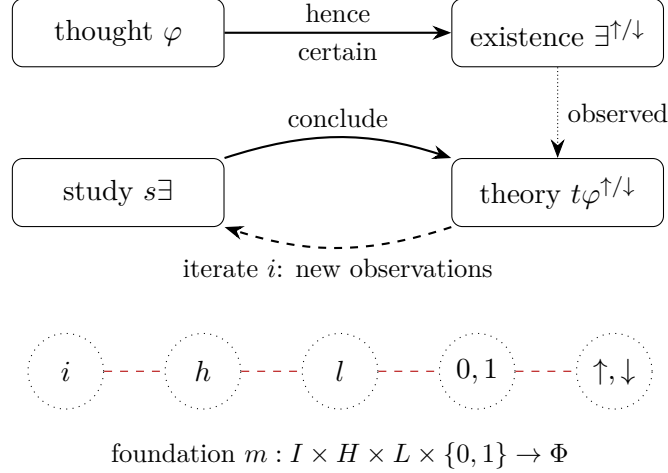


Figure 1: Deduction (top row), induction (middle row) and the foundation (bottom).

Symbol	Informal meaning	Formal reading (Section 3)
φ	thought	proposition $\varphi : \mathcal{W} \rightarrow \text{Prop}$
$\exists$	existence	proposition about worlds; $\mathcal{W}$ is the set of worlds
s	study	list of observations
t	theory	proposition about worlds
$\rightarrow$	hence	semantic entailment $\models$
$\circ$	conclude	support (Definition 3.4)
i	iteration	adding an observation to a study
h, l	hypothesis, level	indices of the foundation map m
$0, 1$	binary state	truth values
$\uparrow, \downarrow$	polarity	affirmation, negation

Table 1: Symbols of the framework.

3 Formal setting

Fix a set $\mathcal{W}$ of *possible worlds*. Every symbol of the diagram is read as follows.

Definition 3.1 (Thought). A *thought* is a proposition about worlds, that is, a map $\varphi : \mathcal{W} \rightarrow \text{Prop}$. We write Φ for the set of thoughts and $\varphi(w)$ for “ φ holds in w ”. An existence is likewise a thought; the distinction between φ and $\exists$ in the diagram is one of role (premise versus conclusion), not of type.

Definition 3.2 (Entailment). A thought φ *entails* ψ , written $\varphi \models \psi$, if $\{w \in \mathcal{W} : \varphi(w)\} \subseteq \{w \in \mathcal{W} : \psi(w)\}$.

Entailment is a relation, not a function: a thought entails many conclusions. This replaces the map $f : \Phi \rightarrow \exists$ of the 2025 version, which could not assign a single conclusion to each thought.

Definition 3.3 (Polarity). The polarities are $\uparrow$ and $\downarrow$. For a proposition p , set $p^\uparrow = p$ and $p^\downarrow = \neg p$.

Definition 3.4 (Study and support). A *study* is a finite list $s = (o_1, \dots, o_n)$ of thoughts, the observations. A theory t is *supported* by s , written $s \circ t$, if some world satisfies t and every o_k :

$$s \circ t \quad :\Longleftrightarrow \quad \exists w \in \mathcal{W}, t(w) \wedge \bigwedge_{k=1}^n o_k(w).$$

Remark 3.5. The 2025 version required $P(T_\varphi \mid S_\exists) > 0$. That quantity is undefined when the study has probability zero. If $\mathcal{W}$ is countable and P gives every world positive probability, then $P(t \wedge s) > 0$ holds exactly when $s \circ t$, so [Definition 3.4](#) is the qualitative content of the original condition and needs no choice of prior. This remark is not part of the Lean formalization.

4 Deduction

Proposition 4.1 (Material implication). *For propositions p, q , $(p \rightarrow q) \iff (\neg p \vee q)$.*
imp_iff_not_or

Proof. If $p \rightarrow q$, then either p holds and so q holds, or $\neg p$ holds; this case split uses the law of excluded middle. Conversely, if $\neg p \vee q$ and p , the first disjunct is impossible, so q . $\square$

Theorem 4.2 (Deduction is certain). *If $\varphi \models \psi$ and $\varphi(w)$, then $\psi(w)$.* *deduction_sound*

Proof. By [Definition 3.2](#), w lies in the set of worlds satisfying φ , which is contained in the set satisfying ψ . $\square$

Proposition 4.3 (Chains of deduction). *If $\varphi \models \psi$ and $\psi \models \chi$, then $\varphi \models \chi$.* *Entails.trans*

Proof. Set inclusion is transitive. $\square$

Proposition 4.4 (Negative polarity is modus tollens). *If $\varphi \models \psi$ and $\psi(w)^\downarrow$, then $\varphi(w)^\downarrow$.*
modus_tollens

Proof. If $\varphi(w)$ held, [Theorem 4.2](#) would give $\psi(w)$, contradicting $\neg\psi(w)$. $\square$

Example 4.5 (Deductive swan). Let B be a set of birds with predicates Swan and White. If $\forall b, \text{Swan}(b) \rightarrow \text{White}(b)$ and $\text{Swan}(a)$, then $\text{White}(a)$. In the diagram this is $\varphi \rightarrow \exists^\uparrow$. *swan_syllogism*

5 Induction

5.1 Iterative refinement

Iteration i adds an observation to a study. The following results say that iteration can only narrow the set of supported theories, which is the precise sense in which induction refines.

Proposition 5.1. *If $(o, o_1, \dots, o_n) \circ t$, then $(o_1, \dots, o_n) \circ t$.* *Supported.of_cons*

Theorem 5.2 (Refinement is monotone). *If every observation of s occurs in s' and $s' \circ t$, then $s \circ t$.* *Supported.mono*

Proof. A world witnessing $s' \circ t$ satisfies t and every observation of s' , hence every observation of s . [Proposition 5.1](#) is the case $s' = (o) ++ s$. $\square$

5.2 Falsification and the problem of induction

For the swan examples take worlds to be colourings $c : B \rightarrow \{\text{true}, \text{false}\}$ recording which birds are white. For a finite list *seen* of birds define

$$\text{SeenWhite}_{\text{seen}}(c) : \Longleftrightarrow \forall x \in \text{seen}, c(x) = \text{true}, \quad \text{AllWhite}(c) : \Longleftrightarrow \forall x \in B, c(x) = \text{true}.$$

Proposition 5.3 (Falsification). *If $\text{Swan}(b)$ and $\neg \text{White}(b)$, then $\neg \forall x, \text{Swan}(x) \rightarrow \text{White}(x)$.*
falsification

Proof. Instantiate the universal statement at b . □

Falsification is deductive: a single observation settles the matter, in line with Popper [9]. Confirmation is not, however many observations are made.

Theorem 5.4 (Induction is not deduction). *Let B have decidable equality, let *seen* be a finite list of birds, and let $b \in B$ with $b \notin \text{seen}$. Then $\text{SeenWhite}_{\text{seen}} \not\models \text{AllWhite}$.* *induction_not_deduction*

Proof. Define $c(x) = \text{false}$ if $x = b$ and $c(x) = \text{true}$ otherwise. Every $x \in \text{seen}$ differs from b , so $c(x) = \text{true}$ and $\text{SeenWhite}_{\text{seen}}(c)$ holds. But $c(b) = \text{false}$, so $\text{AllWhite}(c)$ fails. Hence the worlds satisfying $\text{SeenWhite}_{\text{seen}}$ are not contained in those satisfying AllWhite . □

Proposition 5.5 (Inductive swan). *For every finite list *seen*, $(\text{SeenWhite}_{\text{seen}}) \circlearrowleft \text{AllWhite}$.*
allWhite_supported

Proof. The colouring that makes every bird white satisfies both. □

Together, [Theorem 5.4](#) and [Proposition 5.5](#) state the asymmetry the diagram depicts: the theory “all swans are white” remains supported by every finite study, so it may be held along the circular arrow, but no finite study entails it, so it never travels the straight arrow. This is a formal version of Hume’s problem of induction [4].

6 The foundation

Definition 6.1 (Foundation map). Let I, H, L be sets of iterations, hypotheses and levels, and let $\beta : I \times H \times L \rightarrow \Phi$ assign a base thought to each index. The *foundation map* $m : I \times H \times L \times \{0, 1\} \rightarrow \Phi$ is

$$m(i, h, l, 1) = \beta(i, h, l), \quad m(i, h, l, 0) = \neg \beta(i, h, l).$$

Thus state 1 carries polarity $\uparrow$ and state 0 carries polarity $\downarrow$. In the 2025 version m took values in $\Phi \cup \exists$; since existences are thoughts ([Section 3](#)), the codomain is simply Φ .

Proposition 6.2 (The binary state is the truth value). *For every world w and $b \in \{0, 1\}$, $m(i, h, l, b)(w)$ holds if and only if $\beta(i, h, l)(w)$ agrees with b , that is, $\beta(i, h, l)(w) \Longleftrightarrow b = 1$.*
m_spec

Proof. For $b = 1$ both sides say $\beta(i, h, l)(w)$. For $b = 0$ both sides say $\neg \beta(i, h, l)(w)$. □

7 Discussion

Philosophy. The straight arrow carries Descartes’ movement from thought to existence [3]: what is entailed holds in every world where the premises hold (Theorem 4.2). The circular arrow carries Hume’s and Popper’s view of induction: a generalization is supported, never entailed (Theorem 5.4), and can be overturned by a single counterexample (Proposition 5.3). Levels l in the foundation give a place for Kant’s hierarchical structuring of thought [6].

Education and cognition. The top row of the diagram is suitable for a first course in logic; the foundation invites discussion of polarity and iteration. The visual layout follows the view that diagrams make inferential structure explicit [7, 5].

Limitations and future work.

- Support (Definition 3.4) is qualitative. A quantitative reading of the inductive polarity $t\varphi^\uparrow$ is Bayesian confirmation, $P(t \mid s) > P(t)$; formalizing it requires a probability library such as Mathlib.
- The indices i, h, l of the foundation are uninterpreted sets. Giving levels an order (for example observational below theoretical) would let one state that iteration moves upward.
- Only propositional and first-order statements over a fixed set of worlds are treated. Multi-premise defeasible reasoning and higher-order logic are left open.
- The diagram’s usefulness for learners is an empirical claim that remains to be tested.

8 Conclusion

We have given a diagram for deduction and induction, a possible-worlds semantics for its symbols, and machine-checked theorems for what it depicts: deduction is certain and transitive, negative polarity is modus tollens, iteration narrows the supported theories, one counterexample falsifies, and no finite study entails a universal theory.

A Lean formalization

All results in Sections 4–6 are formalized in Lean 4 (toolchain v4.34.1) in the file `lean/WangLogic/Basic.lean` accompanying this paper. The formalization uses only the core library, contains no `sorry`, and depends at most on the axioms `propext`, `Quot.sound` and `Classical.choice`. It builds with

```
cd lean && lake build
```

The file `lean/ComparatorChallenges/InductionNotDeduction.json` states Theorem 5.4 for independent checking with the Comparator tool. Remark 3.5 is not formalized.

Result	Lean declaration (namespace <code>WangLogic</code>)
Definition 3.2	<code>Entails</code>
Definition 3.4	<code>Supported</code>
Proposition 4.1	<code>imp_iff_not_or</code>
Theorem 4.2	<code>deduction_sound</code>
Proposition 4.4	<code>modus_tollens</code>
Proposition 5.1	<code>Supported.of_cons</code>
Theorem 5.2	<code>Supported.mono</code>
Proposition 5.3	<code>falsification</code>
Theorem 5.4	<code>induction_not_deduction</code>
Proposition 5.5	<code>allWhite_supported</code>
Proposition 6.2	<code>m_spec</code>

References

- [1] Aristotle. Prior analytics, 350 BCE.
- [2] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. Automated Deduction – CADE 28, LNCS 12699, pp. 625–635, 2021.
- [3] René Descartes. Discourse on the method, 1637.
- [4] David Hume. An enquiry concerning human understanding, 1748.
- [5] P. N. Johnson-Laird. *Mental Models: Towards a Cognitive Science of Language, Inference, and Consciousness*. Harvard University Press, 1983.
- [6] Immanuel Kant. Critique of pure reason, 1781.
- [7] Jill H. Larkin and Herbert A. Simon. Why a diagram is (sometimes) worth ten thousand words. *Cognitive Science*, 11(1):65–99, 1987.
- [8] Charles S. Peirce. Prolegomena to an apology for pragmatism. *The Monist*, 16(4):492–546, 1906.
- [9] Karl R. Popper. *The Logic of Scientific Discovery*. Routledge, 1959.